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METABOLIC CONTROL IN ADULTS LIVING WITH TYPE 1 DIABETES ACROSS NORTHWESTERN EUROPE: A COMPARATIVE STUDY

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Background and aims: We have previously identified persistent disparities in glycaemic control and technology use (including continuous glucose monitoring (CGM) and insulin pumps) among people living with type 1 diabetes (T1D) across populations and age groups. However, comparing metabolic control between countries has been

challenging due to differences in reported treatment targets and variations in data sources, whether clinical or national registries. Our aim is to present an updated comparison of metabolic control in adults living with T1D across five European countries, providing insights to support strategies for improving diabetes management and minimizing variability between countries.

Materials and methods: We analyzed data from 171 431 adults living with T1D (aged 18-79 years), covering the previous 12-24 months (data 2021-2023), sourced from population-based national registers and quality improvement initiatives in five countries: Belgium (BE), Finland (FI), Norway (NO), Scotland (SC), and Sweden (SW). We compared distributions of HbA_{1c}, systolic blood pressure and LDL cholesterol, alongside the use of continuous subcutaneous insulin infusion (CSII), CGM, and statins across the countries. The relation between CSII use and HbA_{1c} was assessed using Pearson correlation.

Results: The study population varied from 22 283 to 50 476 between countries. The distribution of metabolic indicators and treatments is shown in Table 1. The proportion of patients with HbA_{1c} ≤53 mmol/mol (≤7%) varied between 43% (SW) and 18% (SC). The proportion with systolic blood pressure ≤140 mmHg varied between 86% (SW) and 57% (SC), whereas LDL cholesterol ≤2.5 mmol/L varied between 77% (BE) and 35% (SC). Countries with higher proportions of CSII users had higher proportions of people with HbA_{1c} ≤53 mmol/mol (≤7%).

Conclusion: This study highlights notable variations in metabolic control, including blood pressure and blood lipid control, and the use of diabetes technology across five European countries, underscoring the need for tailored strategies to improve diabetes management. Countries with greater use of CSII showed better glycaemic control, suggesting that increased access to advanced diabetes technologies may help reduce disparities. However, further research is required. These findings emphasize the importance of aligning treatment goals and healthcare strategies across nations to minimize variability in outcomes and optimize diabetes care for all people living with T1D.

Table 1: Metabolic indicators and treatment of the study population (T1D, aged 18-79 years), by country

		Belgium	Finland	Norway	Scotland	Sweden
Number of individuals		33 630	37 333	22 283	27 709	50 476
HbA_{1c}	Recorded, %	98	94	97	87	97
	≤53 mmol/mol (≤7%), %	35	30	40	18	43
Systolic blood pressure	Recorded, %	97	51	82	80	89
	≤140 mmHg, %	79	N/A	80	57	86
LDL cholesterol	Recorded, %	91	91	94	54	85
	≤2.5 mmol/L, %	77	73	58	35	64
CSII use, %		18	N/A	41	16	33
CGM use, %		85	N/A	90	N/A	90
Statin use, %		44	58	42	57	46

CSII: continuous subcutaneous insulin infusion; CGM: continuous glucose monitoring; N/A: data not available